

Information theory and bulk rotational or vibrational relaxation processes

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The possible occurrence in bulk relaxation of pure-exponential, near-exponential, and non-exponential decay of the total energy is examined in terms of the normal-mode and information-theoretic approaches. Experimental tests are suggested for the identification of pure-exponential decay caused by adherence to the 'sum rule', and of near-exponential decay. In the case of near-exponential decay (as opposed to pure-exponential decay), it is not possible to derive reliable state-to-state rate constants by invoking approximate adherence to the sum rule.

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On examine la possibilité de l'existence, dans la relaxation globale, de décompositions exponentielles pures, pratiquement exponentielle et non-exponentielle de l'énergie totale en termes des approches du mode normal et de l'information théorique. On suggère des essais expérimentaux permettant d'identifier une décomposition exponentielle pure par une adhésion à la "règle des sommes" ainsi qu'une décomposition pratiquement exponentielle. Dans le cas d'une décomposition pratiquement exponentielle (par opposition à une décomposition exponentielle pure), il n'est pas possible d'obtenir des constantes de vitesse d'état à état qui soient certaines en invoquant l'adhésion approximative à la "règle des sommes".

[Traduit par le journal]

Two recent papers from this laboratory have examined the bulk relaxation behaviour of a model diatomic molecule, infinitely diluted in an inert-gas bath, in terms of normal modes of relaxation (1, 2). An alternative treatment of simple bulk relaxation processes, in terms of an information-theoretic approach (3), has also been developed recently by Procaccia and Levine (4-6). On the surface, it appears that these two treatments are not always consistent with each other: in particular, markedly non-exponential decay of the total energy would seem to be a distinct possibility, given the right kind of experiment (2), whereas accepting pure-exponential decay as a constraint, the information-theoretic treatment provides an attractive and elegant framework within which to describe such bulk relaxation processes. Since both approaches to the problem are based on the application of

the master equation, we will begin by restating the relevant useful equations.

The isothermal relaxation of a diatomic gas, with N rotation-vibration levels, diluted in a heat bath is governed by the transition matrix $\mathbf{A}$ where (1)

$$[1] \quad A_{ij} = [\mathbf{M}]\mathbf{Z} \left\{ (1 - \delta_{ij})P_{ji} - \delta_{ij} \times \sum_{k=0}^{N-1} (1 - \delta_{ik})P_{ki} \right\}$$

In [1], P_{ji} is the probability per collision for a transition from state i to state j , $[\mathbf{M}]$ is the concentration of buffer gas, and $\mathbf{Z}$ is the collision number. As a consequence of detailed balancing

$$[2] \quad A_{ij} \tilde{n}_i = A_{ji} \tilde{n}_j$$

where $\tilde{n}_i$ is the population of the i th level at equilibrium, $\mathbf{A}$ may be symmetrized to $\mathbf{B}$ by the

transformation

$$[3] \quad \mathbf{B} = \mathbf{N}^{1/2} \mathbf{A} \mathbf{N}^{-1/2}$$

where

$$[3a] \quad N_{ij} = \delta_{ij} \tilde{n}_i$$

The matrix $\mathbf{B}$ therefore has real eigenvalues, λ_j , which may be found using the transformation matrix $\mathbf{S}$, *i.e.*

$$[4] \quad \mathbf{B} \mathbf{S} = \mathbf{S} \mathbf{A}$$

The final attainment of Boltzmann equilibrium is guaranteed by the fact that one eigenvalue, λ_{N-1} , is zero, with a corresponding eigenvector

$$[4a] \quad S_{i,N-1} = \tilde{n}_i^{1/2}$$

and all of the other eigenvalues are negative. The time evolution of the total internal energy of the system, $E(t)$, is

$$[5] \quad E(t) = \sum_{i=0}^{N-1} \epsilon_i n_i(t) \\ = \sum_{j=0}^{N-1} e^{\lambda_j t} \sum_{i=0}^{N-1} \tilde{n}_i^{1/2} \epsilon_i S_{ij} \\ \times \sum_{k=0}^{N-1} S_{kj} \tilde{n}_k^{-1/2} n_k(0)$$

where $n_i(t)$ is the population of the i th level at time t , and ϵ_i is the energy of that level. An alternative way of writing [5] is (1)

$$[5a] \quad E(t) = \sum_{j=0}^{N-1} \xi_j e^{\lambda_j t}$$

where ξ_j , the amount of energy which decays with characteristic time constant λ_j , is given by

$$[6] \quad \xi_j = \sum_{i=0}^{N-1} \tilde{n}_i^{1/2} \epsilon_i S_{ij} \sum_{k=0}^{N-1} S_{kj} \tilde{n}_k^{-1/2} n_k(0)$$

Inspection of [5] or [5a] shows that, in general, the energy must decay in a non-exponential manner: pure-exponential decay will occur under rather special circumstances. Nevertheless, these model calculations (1, 2) have shown that, often, very few of the normal modes may have associated with them a significant value of ξ_j , and in these cases the relaxation behaviour would be very difficult indeed to distinguish from a pure-exponential decay under normal experimental conditions; other examples are given by Cukier and Hynes (7).

Pure-exponential decay can occur in two ways. First, in a trivial way, when all the non-zero

eigenvalues are degenerate: the necessary and sufficient condition for this to occur (8) is that the transition rate matrix should have the form

$$[7] \quad A_{ij} = \alpha \left\{ (1 - \delta_{ij}) \tilde{n}_j - \delta_{ij} \times \sum_{k=0}^{N-1} (1 - \delta_{ik}) \tilde{n}_k \right\}$$

where α is a constant.¹ Second, a much less restrictive and well-known condition (4-13), that the transition rates obey the 'sum rule'

$$[8] \quad \sum_j \epsilon_j A_{ij} = \mu(E(\infty) - \epsilon_i)$$

for all i , where μ is a positive constant, whence

$$[8a] \quad \frac{dE(t)}{dt} = \mu[E(\infty) - E(t)]$$

Equation 7 is, in fact, a special case of [8]. In conventional terms, we have $0 = \lambda_{N-1} > -\mu = \lambda_{N-2} \geq \lambda_{N-3} \geq \dots \lambda_0$; adapting the formula of van Kampen (13), the vector of the symmetrized form of [8] corresponding to the eigenvalue $-\mu$ is

$$[9] \quad S_{i,N-2} = \frac{\tilde{n}_i^{1/2}(E(\infty) - \epsilon_i)}{\{E^2(\infty) - (E(\infty))^2\}^{1/2}}$$

Using this, together with [4a], it is then a simple matter (8) to show that all the other modes of relaxation are passive in the sense that all ξ_j , $j < (N-2)$ are zero: hence each of these $(N-2)$ modes only shuffles the populations without changing the total energy of the assembly of molecules.

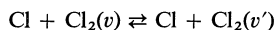
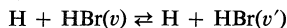
The occurrence of pure-exponential decay requires certain relationships between the transition rates. To begin with, we may classify possible transition-probability patterns into three distinct classes.

(i) Those which conform to eq. 7, whence both the energy and all the individual populations decay exponentially: we do not know whether there exist any examples of such behaviour in molecular relaxation.

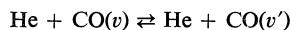
(ii) Those which conform to the sum rule, eq. 8, whence the energy will decay exponentially, but the populations will not. Several examples of such conformity are known (7), most notably the Landau-Teller transition probabilities for harmonic-oscillator transitions. It appears also

¹Since this paper was submitted, we have learnt that this condition has been proved also by Procaccia *et al.* (9).

that certain classically computed V-T transition probabilities (14) for



conform approximately to the sum rule (5). On the other hand, some transition probabilities derived from quantum calculations, in the breathing-sphere approximation (15), on



do not conform, in the sense that the energy evolution derived from these probabilities is not exponential for some initial population distributions.

Adherence to the sum rule is verifiable, in principle at least, by experiment since at long time, after all the initial transients have died out, the evolution of the populations depends on the initial energy but is independent of the fine structure of the initial distribution of that energy. Thus (8)

$$[10] \quad \begin{aligned} n_i(t) &\rightarrow \tilde{n}_i + n_i(N-2, t) \\ (t \gg -\lambda_{N-3}^{-1}) & \\ &= \tilde{n}_i + \tilde{n}_i e^{-\mu t} \frac{[E(\infty) - \varepsilon_i][E(\infty) - E(0)]}{\{E^2(\infty) - (E(\infty))^2\}} \end{aligned}$$

(iii) Those which conform to neither [7] nor [8], in which case neither the energy nor the populations decay in a strictly exponential fashion. Again, we may subdivide into three classes: (a) those probabilities which conform to linear surprisal,² in which case, as is shown below, the range of eigenvalues can be very small; (b) those which conform to more complex surprisal forms (9, 16), a development which arose after we had completed this analysis, and which we will not discuss further; (c) arbitrary transition probabilities, such as those used in our own previous model calculations (1, 2), where the range of the eigenvalues can be very large indeed, *cf.* also ref. 17, Fig. 3.

The 'linear surprisal' form of the rate constant is

$$[11] \quad k_{i \rightarrow j} = k_{i \rightarrow j}^0 \exp(-\theta|\varepsilon_j - \varepsilon_i|/kT) \equiv ZP_{ji}$$

where $k_{i \rightarrow j}^0$ is the 'prior rate constant' (the rate constant in the absence of any dynamical constraint) and θ is the 'surprisal' (a measure of the

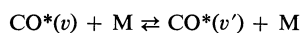
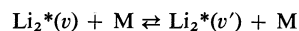
effect of the energy defect for transition from state i to state j). For a system with an array of rotation-vibration levels at temperature T (18)

$$[12] \quad k_{vJ \rightarrow v'J'}^0 = \text{constant} \times T^{1/2} g_{J'} \times \exp(\Delta) \cdot \Delta \cdot K_1(\Delta)$$

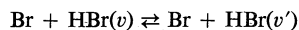
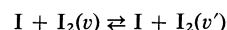
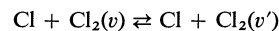
where $g_{J'} = 2J' + 1$, $\Delta =$ 'reduced energy transfer' $= (\varepsilon_{vJ} - \varepsilon_{v'J'})/2kT$, and $K_1(\Delta) =$ modified Bessel function of the first order. Since $K_1(\Delta)$ is odd with respect to Δ , eq. 12 satisfies the principle of detailed balancing. This implies that the surprisal θ in eq. 11 is symmetric with respect to the initial and final states, and the rate constant then takes the form

$$[13] \quad k_{vJ \rightarrow v'J'} = \text{constant} \times (2J' + 1) \times \exp(\Delta) \cdot \Delta \cdot K_1(\Delta) \cdot \exp(-2\theta|\Delta|)$$

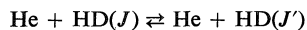
As examples of V-T transfer, the experimental rates of



have been shown to conform to a linear surprisal form (4). Among the theoretically calculated rates (14)



have been shown to obey (approximately) the linear surprisal form, although the last reaction does not obey the sum rule (5, 6); more recently (9, 16) the $\text{Cl} + \text{Cl}_2$ case has been re-analysed, giving a functional (v, v') dependence on the surprisal. No quantal V-T data have yet, to our knowledge, been analysed in this way, but we have shown (8) that the impulsive limit of the first-order distorted-wave approximation for nearest-neighbour transitions has the same temperature dependence as the linear surprisal form. For R-T collisions, the quantal data of Green (19) on



have been analysed by Procaccia and Levine (20): there is (semiquantitative) agreement in respect of the temperature coefficients, and the fit to the linear surprisal form is accurate to about $\pm 20\%$.

Given that many transition probabilities appear to obey the linear surprisal form, it is a simple matter to show that, at least for V-T and

²Linear surprisal forms may or may not obey the sum rule.

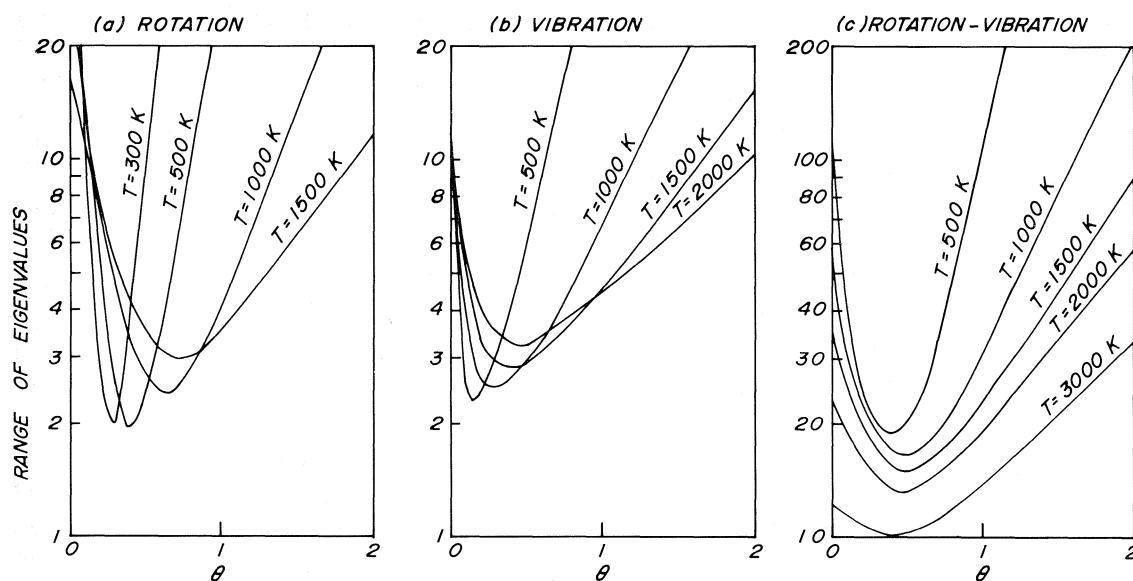


FIG. 1. Comparison of the range of eigenvalues λ_0/λ_{N-2} for the information-theoretic descriptions of three model molecules (see text) for a series of temperatures and surprisal values: (a) pure rotational relaxation; (b) pure vibrational relaxation; (c) mixed rotation-vibration relaxation.

R-T processes, one would expect to find *almost* pure-exponential decay of the energy, *even if* the sum rule is not obeyed. Since in eq. 13, the 'reduced energy transfer' is representative for all diatomic molecules in either pure vibrational or pure rotational relaxation if a reduced temperature scale is used, *viz.* $\beta_v = hc\omega_e/kT$ or $\beta_R = hcB_e/kT$, we can continue to use our previous model molecule, *ortho*-H₂, in this illustration. Figure 1 shows plots of the ratio λ_0/λ_{N-2} (*i.e.* the spread of the eigenvalues) for a range of values of θ , for three kinds of processes: (a) pure rotational relaxation involving the lowest 8 rotational states of *ortho*-H₂; (b) pure vibrational relaxation involving the lowest 8 vibrational states of $J=1$ H₂; ³ (c) mixed vibration-rotation relaxation involving the lowest 20 levels of *ortho*-H₂. Over a wide range of reduced temperature, the minimum spread in eigenvalues occurs near $\theta = 0.5$, compared with values of θ of the order of unity found in many analyses reported in the literature (20). The fact that the spread of eigenvalues is indeed modest means that these *distinct* relaxation times will be hard to resolve experimentally (1,

2) and the bulk relaxation will *appear* to be very close to a pure-exponential decay, because, in fact, we have an approximation to the $(N-1)$ -fold degenerate case, eq. 7.

However, as is well known (1, 7), near-exponential decay (we may define near-exponential decay to mean that it would be difficult to detect experimentally any non-exponential behaviour) can occur when the eigenvalues are well separated, and this is shown by two pairs of calculations illustrated in Fig. 2. In these calculations, we compare the bulk relaxation behaviour for (a) rotation and (b) vibration, for a temperature change from 700 to 500 K, using transition probabilities *either* of the functional form of [13] *and* having the minimum spread of eigenvalues (see Fig. 1) *or* the empirical set of transition probabilities used previously (1, 2). The lower curve shows that the behaviour is hardly distinguishable⁴ for bulk vibrational relaxation, despite the fact that in one case the spread of eigenvalues λ_0/λ_{N-2} is 2.3, whereas in the other case it is 120; in normal-mode terms, this is because the succession of normal modes of longest time constant (see *e.g.* Fig. 3 of ref. 2) are relatively close together, with the last one carrying most of the traffic (*i.e.* ξ_{N-2} dominates).

³In the usual formulation, each vibrational state is averaged over its rotational manifold, assuming rotational equilibrium (5). However, there is some difficulty with detailed balancing in such a formulation, and since we are interested at the present time in only a model calculation, we consider only one rotational state.

⁴Of course, drawn out on a much larger scale, both lines are slightly curved.

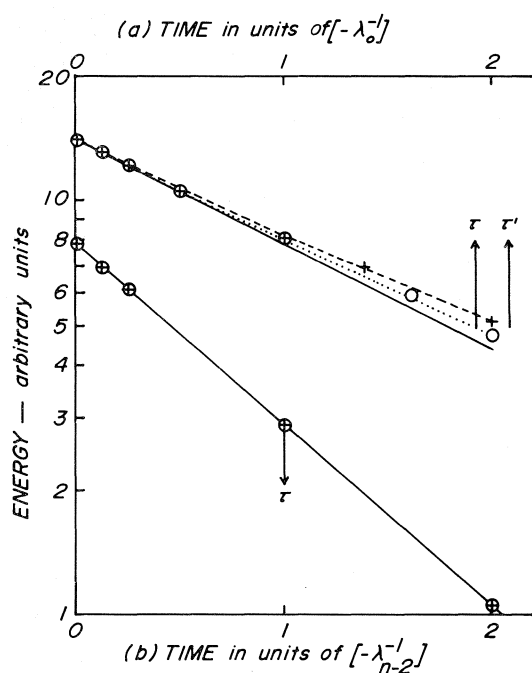


FIG. 2. Logarithmic plots of the total energy *vs.* time for (a) pure rotational relaxation and (b) pure vibrational relaxation. Open circles represent points calculated using information-theoretic rate constants of *minimum eigenvalue spread*, and crosses represent points calculated with an empirical set of transition rates (1). The solid lines represent exact exponential decay, and the vertical arrows marked τ represent the points at which $1/e$ of the total energy remains: this occurs very close to $-\lambda_{N-2}^{-1}$ in the vibrational cases, but near $-2\lambda_0^{-1}$ in the rotational cases. The eigenvalue ranges (*i.e.* λ_0/λ_{N-2}) in these calculations were (a) rotation 1.9 and 75, (b) vibration 2.3 and 120 for the information-theoretic and the empirical rates respectively.

Non-exponential decay of vibrational energy will always be difficult to detect unless the molecule is very anharmonic (which will cause a wider spacing of the eigenvalues) and the relative populations of high- v and low- v states change appreciably in the relaxation process.

The upper part of Fig. 2 shows the rotational relaxation behaviour using probabilities which have an eigenvalue range λ_0/λ_{N-2} of 1.9 and 75 respectively. Here the plots are just about distinguishable from each other, and they are both slightly curved. The curvature is of course more severe for the case where the range of eigenvalues is larger, and it is interesting to note that, in this case, the time it takes the energy to relax is closely related to the *fastest* time constant, not the slowest as is usually assumed: Cukier and

Hynes (7) also give a numerical example of this kind, and the reasons for this behaviour are easily understood in terms of our previous discussion of rotational sound dispersion (1). For these reasons also, non-exponentiality should be much easier to detect in rotational than in vibrational relaxation, and a relatively simple experiment based on our earlier calculations (2) should suffice to settle the issue one way or the other: all that is necessary is to measure the shock-wave rotational relaxation time for hydrogen from a series of different starting temperatures to a single fixed final temperature; if a monotonically increasing progression of relaxation times is found, converging to the ultrasonic relaxation time for the final temperature in question, then the decay is non-exponential and the treatment depicted in Fig. 1 of ref. 2 would be essentially correct. In the case of T-VR processes, we can expect, as discussed previously (2), a complete spectrum of behaviour from that in which the rotational and vibrational relaxations separate completely to that where they are completely inseparable: in the former case, the considerations we have just discussed would apply, whereas in the latter case, noting (a) the larger spread of eigenvalues shown in Fig. 1c and (b) the apparent failure of Alexander's calculations (21) for T-VR processes in hydrogen to conform to the linear surprisal form, it would seem that non-exponential decay will be the rule rather than the exception. Thus, in fact, we see no real and irreconcilable differences between the normal-mode and the information-theoretic descriptions of bulk relaxation processes.

However, there is one aspect of the information-theoretic approach where we feel that some caution is appropriate. Procaccia and Levine (4-6) have claimed that if the observed decay is exponential, then it is possible to deduce microscopic state-to-state transition rates from the observed *bulk relaxation* time constant. If the sum rule holds, this may well be true, although some doubts have been expressed⁵ about this possibility unless the relaxing system behaves in a canonically invariant manner. But it seems that adherence to the sum rule, from the evidence cited above, is unlikely to be the general behaviour, in which case the claim can have, at best, only limited validity. There are two difficulties,

⁵K. E. Shuler and R. I. Cukier, personal communications.

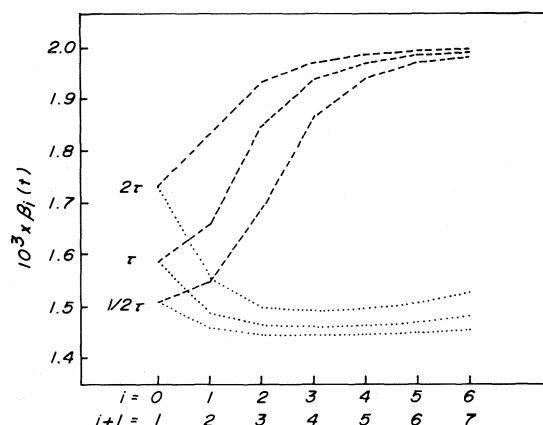


FIG. 3. Comparison of vibrational temperatures at various levels for a sudden cooling process from 700 to 500 K. The temperature is defined as $\beta_i = k \ln [n_i(t)/n_{i+1}(t)]/(\epsilon_{i+1} - \epsilon_i)$, and the plots are shown for $t = \frac{1}{2}\tau$, τ , and 2τ where $\tau = -\lambda_{N-2}^{-1}$. Dashed line = empirical probabilities (1); dotted line = information-theoretic probabilities of minimum eigenvalue spread.

in our view. First, a relaxation may appear to be exponential, or very nearly so, but this in no way implies that the sum rule holds, unless the long-time behaviour of the populations is shown to conform to eq. 10. Second, as can be seen from Fig. 2 (especially in the vibrational case), the bulk relaxation behaviour is extremely insensitive to the transition probabilities assumed: hence, as Carrington (22) recognized, it is a very ill-conditioned problem to work back from the observed bulk relaxation behaviour to the detailed rate constants. As an example of this, we show in Fig. 3 the behaviour of the vibrational population distributions at $\frac{1}{2}\tau$, τ , and 2τ for the two relaxations which are energetically indistinguishable in Fig. 2. Although the bulk relaxation behaviours are only imperceptibly different, the populations show diametrically different evolution behaviours, and it is evident that the ill-conditioned nature of the problem is quite severe: we can do no more than reiterate Carrington's conclusion of 15 years ago, that the present precision of relaxation experiments does not allow a meaningful deduction of transition rate constants from observed bulk relaxation

rates. Nevertheless, in the absence of other sources of information, the information-theoretic approach (9) still provides the least-biased guess at a set of transition rates which will reproduce the known behaviour.

Acknowledgments

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